

Lecture 3: Two-Sample T-Test and ANOVA

MDR Notes

1 Two-Sample T-Test and Regression

The 2-sample t-test is equivalent to testing for $\hat{\beta}_1 = 0$ in regression model with 2 levels (categories) of one factor.

- Group 1: $x_1, \dots, x_m$
- Group 2: $y_1, y_2, \dots, y_n$

1.1 Pooled Variance(HW of last class)

The residuals ($\hat{e}$) are the deviations from each group's mean:

$$\hat{e}_i = \begin{cases} x_i - \bar{x} & \text{for } i = 1, \dots, m \\ y_j - \bar{y} & \text{for } j = 1, \dots, n \end{cases}$$

The pooled variance, s_p^2 , is the sum of squared residuals divided by the degrees of freedom:

$$s_p^2 = \frac{\sum \hat{e}_i^2}{m + n - 2} = \frac{\sum_{i=1}^m (x_i - \bar{x})^2 + \sum_{j=1}^n (y_j - \bar{y})^2}{m + n - 2}$$

2 ANOVA (Analysis of Variance)

This section covers one factor with multiple levels. ANOVA is a test for the equality of means (not variance).

2.1 Sum of Squares Decomposition

Let the grand mean be $\bar{Y}_{GM} = \frac{m\bar{x} + n\bar{y}}{m+n}$.

2.1.1 Total Sum of Squares (SST)

$$SST = \sum_{i=1}^m (x_i - \bar{Y}_{GM})^2 + \sum_{j=1}^n (y_j - \bar{Y}_{GM})^2$$

2.1.2 Sum of Squares Between (SSB) / Regression (SSReg)

This is the variation between the groups.

$$SSReg(SSB) = \frac{mn}{m+n}(\bar{x} - \bar{y})^2 \quad (1)$$

$$= m(\bar{x} - \bar{Y}_{GM})^2 + n(\bar{y} - \bar{Y}_{GM})^2 \quad (2)$$

2.1.3 Sum of Squares Within (SSW) / Error (SSR)

This is the variation within the groups (the same as the numerator for pooled variance).

$$SSR(SSW) = \sum_{i=1}^m (x_i - \bar{x})^2 + \sum_{j=1}^n (y_j - \bar{y})^2 = \sum \hat{e}_i^2$$

2.1.4 Fundamental ANOVA Identity

$$SST = SSReg + SSR$$

2.2 F-Statistic

The F-statistic compares the between-group variation to the within-group variation.

$$F = \frac{SSB/(2-1)}{SSW/(m+n-2)} = \frac{SSB/1}{SSE/(m+n-2)}$$

Interpretation:

- The variation **b/w** groups is much larger than the variation **within** the groups (i.e., F is large),
- $\Rightarrow$ we conclude the group are significantly separated.

3 One-Way ANOVA Model

For one factor with J levels, where l_i is the number of observations in level i :

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij}, \quad \text{for } i = 1, \dots, I, \quad j = 1, \dots, l_i$$

Where:

- $E[y_{ij}] = \mu + \alpha_i$ (This is the mean of the i^{th} group)
- μ is the overall mean
- $\mu + \alpha_i$ is mean of particular group

3.1 Hypothesis

- $H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_I = 0$ (All group means are equal)
- $H_a : \text{not } H_0$ (i.e., at least one α_i is different)

3.2 Identifiability Problem

This model is **not identifiable**. That is, there are at least 2 distinct sets of parameters that give the same probability distribution for y_{ij} . The model has too many parameters.

Example: For $I = 2$ levels, the parameters are $(\mu, \alpha_1, \alpha_2)$.

$$\begin{pmatrix} \mu \\ \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \quad \text{gives the same model as} \quad \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

In both cases, the group means are:

$$\begin{pmatrix} \mu + \alpha_1 \\ \mu + \alpha_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

3.2.1 Model in Matrix Form

The model can be written using indicator variables:

$$y_i = \mu + \alpha_1 I(x_i = 1) + \alpha_2 I(x_i = 2) + \dots + \alpha_I I(x_i = I)$$

$$Y = X\beta + \epsilon \quad \text{where} \quad \beta = \begin{bmatrix} \mu \\ \alpha_1 \\ \dots \\ \alpha_I \end{bmatrix}$$

An example design matrix X for $I = 3$ levels (with $l_1 = 2, l_2 = 3, l_3 = 1$):

$$X = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}_{n \times (I+1)}$$

Here, we have $I + 1 = 4$ parameters, but $\text{Rank}(X) = I = 3$. There is one extra parameter. We need to remove dim-rank, i.e. 1 number of parameters.

3.2.2 Constraints for Identifiability

To make the model identifiable, we must add one constraint.

1. **Set $\mu = 0$**
 - α_i directly represents the mean of the i^{th} level.
2. **Set $\alpha_1 = 0$ (Reference level)**
 - μ represents the mean of level 1.
 - $\alpha_i = (\text{mean of level } i) - (\text{mean of level } 1) = (\mu + \alpha_i) - (\mu)$ for $i \neq 1$.
3. **Sum-to-zero: $\sum_{i=1}^I \alpha_i = 0$**
 - μ represents the overall grand mean.
 - α_i represents the i^{th} level's deviation from the overall mean.

The choice of constraint depends on the desired interpretation of the estimated parameters.